

## **Engineering Sprint 8 – Designing Asynchronous Workflows**

### **Objective**

The objective of Sprint 8 is to understand asynchronous processing in Salesforce and learn how to decide when work should be executed immediately and when it should be processed in the background. The sprint introduces Future Methods, Queueable Apex, Batch Apex, and Scheduled Apex, emphasizing that the choice depends on business requirements rather than simply knowing the technology.

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### **Synchronous vs Asynchronous Processing**

#### **Synchronous Processing**

In synchronous processing, the user waits until all business logic is completed before receiving a response. This approach is suitable for operations where immediate feedback is required, such as validating a student's eligibility before submitting an application.

#### **Asynchronous Processing**

In asynchronous processing, only the essential work is completed immediately, while secondary tasks are executed later in the background. This improves system responsiveness because the user does not have to wait for non-critical operations.

### **Asynchronous Apex Technologies**

#### **Queueable Apex**

Queueable Apex represents a structured background job that Salesforce places in an execution queue. It is useful when background processing involves multiple related activities or when a sequence of jobs is required. For new asynchronous work, Queueable Apex is often the preferred approach because it provides a richer job-oriented model.

Developer Console - Google Chrome

curious-raccoon-5fz8-dev-ed.trilblaze.my.salesforce.com/\_ui/common/apex/debug/ApexCSIPage

File • Edit • Debug • Test • Workspace • Help • < • >

OfferPostProcessingJob.apex • Log executeAnonymous @8/7/2026, 10:41:43 AM

**Execution Log**

| Timestamp    | Event      | Details                                   |
|--------------|------------|-------------------------------------------|
| 10:41:43:009 | USER_DEBUG | [11]DEBUG[Queueable Job Started           |
| 10:41:43:009 | USER_DEBUG | [13]DEBUG[Offer ID : 00100000000000001AAA |
| 10:41:43:009 | USER_DEBUG | [15]DEBUG[Background Processing Completed |

☐ This Frame ☐ Executable ☐ Debug Only ☒ Filter USER\_DEBUG

Logs Tests Checkpoints Query Editor View State Progress Problems

| User                          | Application | Operation                                 | Time                  | Status  | Read   | Size    |
|-------------------------------|-------------|-------------------------------------------|-----------------------|---------|--------|---------|
| PADALA DURGA SETHA PRAVALLIKA | Unknown     | QueueableHandler                          | 8/7/2026, 10:41:43 AM | Success |        | 3.26 KB |
| PADALA DURGA SETHA PRAVALLIKA | Unknown     | /services/data/v67.0/tooling/executeAn... | 8/7/2026, 10:41:42 AM | Success | Unread | 3.7 KB  |

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## 2. Future Method

Future Methods execute eligible Apex code asynchronously. They are suitable for simple background processing, such as notifying an external system after an application has been saved. The recommended approach is to pass only the required record identifiers and retrieve the latest data during execution.

Developer Console - Google Chrome

curious-raccoon-5fz8-dev-ed.trilblaze.my.salesforce.com/\_ui/common/apex/debug/ApexCSIPage

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OfferPostProcessingJob.apex • Log executeAnonymous @8/7/2026, 10:41:43 AM • FutureSample.apex • Log executeAnonymous @8/7/2026, 10:44:31 AM

**Execution Log**

| Timestamp    | Event      | Details                        |
|--------------|------------|--------------------------------|
| 10:44:31:031 | USER_DEBUG | [7]DEBUG[Future Method Running |

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Logs Tests Checkpoints Query Editor View State Progress Problems

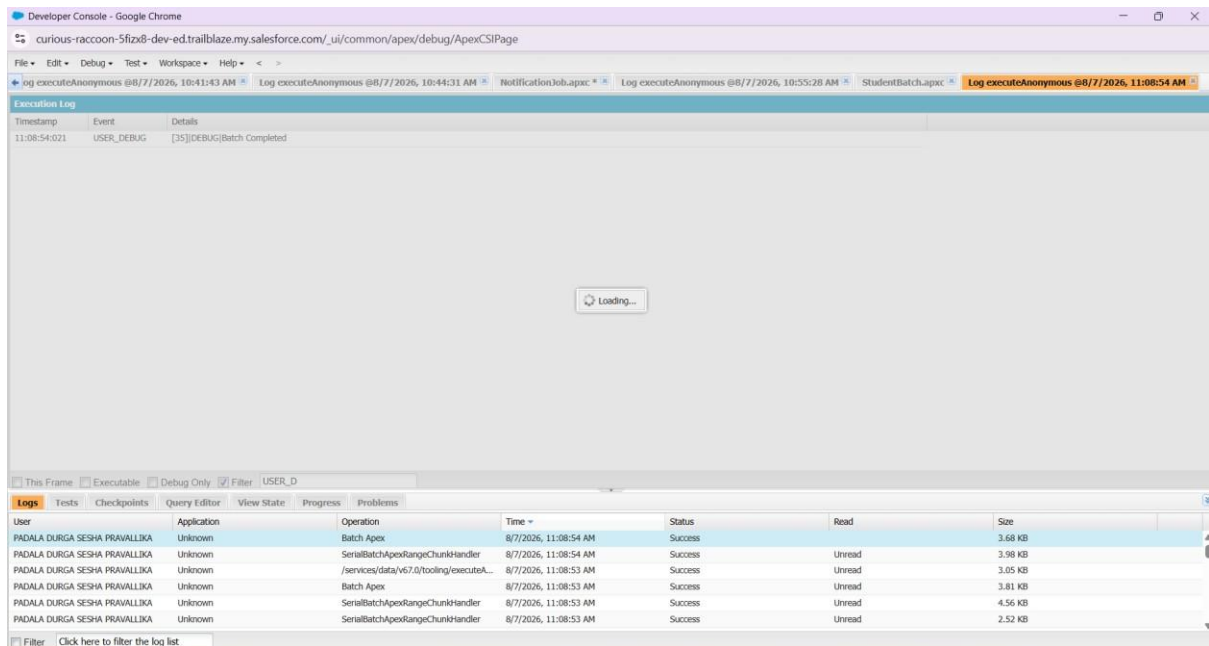
| User                          | Application | Operation                                 | Time                  | Status  | Read   | Size    |
|-------------------------------|-------------|-------------------------------------------|-----------------------|---------|--------|---------|
| PADALA DURGA SETHA PRAVALLIKA | Unknown     | FutureHandler                             | 8/7/2026, 10:44:31 AM | Success |        | 2.82 KB |
| PADALA DURGA SETHA PRAVALLIKA | Unknown     | /services/data/v67.0/tooling/executeAn... | 8/7/2026, 10:44:30 AM | Success | Unread | 2.05 KB |
| PADALA DURGA SETHA PRAVALLIKA | Unknown     | QueueableHandler                          | 8/7/2026, 10:41:43 AM | Success |        | 3.26 KB |
| PADALA DURGA SETHA PRAVALLIKA | Unknown     | /services/data/v67.0/tooling/executeAn... | 8/7/2026, 10:41:42 AM | Success | Unread | 3.7 KB  |

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30°C Mostly cloudy Search 10:45 07-08-2026

## Batch Apex

**Batch Apex is designed for processing very large datasets. Instead of handling all records in a single transaction, the work is divided into smaller batches. A Batch Apex class typically contains the start, execute, and finish stages to manage the overall processing flow.**



The screenshot shows the Salesforce Developer Console interface. The top section displays the 'Execution Log' with a table of events. The bottom section shows the 'Logs' tab with a detailed log table.

| Timestamp    | Event      | Details                   |
|--------------|------------|---------------------------|
| 11:08:54:021 | USER_DEBUG | [35]DEBUG:Batch Completed |

| User                          | Application | Operation                                | Time                  | Status  | Read   | Size    |
|-------------------------------|-------------|------------------------------------------|-----------------------|---------|--------|---------|
| PADALA DURGA SETHA PRAVALLIKA | Unknown     | Batch Apex                               | 8/7/2026, 11:08:54 AM | Success |        | 3.68 KB |
| PADALA DURGA SETHA PRAVALLIKA | Unknown     | SerialBatchApexRangeChunkHandler         | 8/7/2026, 11:08:54 AM | Success | Unread | 3.98 KB |
| PADALA DURGA SETHA PRAVALLIKA | Unknown     | /services/data/v67.0/tooling/executeA... | 8/7/2026, 11:08:53 AM | Success | Unread | 3.05 KB |
| PADALA DURGA SETHA PRAVALLIKA | Unknown     | Batch Apex                               | 8/7/2026, 11:08:53 AM | Success | Unread | 3.81 KB |
| PADALA DURGA SETHA PRAVALLIKA | Unknown     | SerialBatchApexRangeChunkHandler         | 8/7/2026, 11:08:53 AM | Success | Unread | 4.56 KB |
| PADALA DURGA SETHA PRAVALLIKA | Unknown     | SerialBatchApexRangeChunkHandler         | 8/7/2026, 11:08:53 AM | Success | Unread | 2.52 KB |

## Scheduled Apex

Scheduled Apex allows business logic to run automatically at a specified time without user interaction. It is suitable for recurring operations such as processing expired job opportunities every morning.

Engineering Sprint 8 StepsHands-On OrgsScheduled Jobs | SalesforceAsk Gemini

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Developer Edition

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Didn't find what you're looking for? Try using Global Search.

SETUP

Scheduled Jobs

Help for this Page

All Scheduled Jobs

The All Scheduled Jobs page lists all of the jobs scheduled by your users. Multiple job types may display on this page. You can delete scheduled jobs if you have the permission to do so.

Percentage of Scheduled Jobs Used: 1%  
You have currently used 1 scheduled Apex jobs out of an allowed organization limit of 100 active or scheduled jobs. To learn about how this limit is calculated and what contributes to it see the Lightning Platform Apex Limits topic.

View: All Scheduled Jobs

Create New View

Schedule Apex

| Action                   | Job Name ↑                                          | Submitted By                    | Submitted            | Started             | Next Scheduled Run   | Type                       | Cron Trigger ID |
|--------------------------|-----------------------------------------------------|---------------------------------|----------------------|---------------------|----------------------|----------------------------|-----------------|
| Manage   Del   Pause Job | Daily Job                                           | DURGGA SIESHA PRAVALLIKA PADALA | 07/08/2026, 11:10 am |                     | 08/08/2026, 10:00 am | Scheduled Apex             | 08ehg000000qFY  |
| Del                      | Metalytics Data Loader Job for Org : 000hg000002hg7 | User Integration                | 28/07/2026, 10:08 am | 07/08/2026, 5:07 am | 08/08/2026, 5:07 am  | Autonomous Data Loader Job | 08ehg000000cL13 |

A B C D E F G H I J K L M N O P Q R S T U V W X Y Z Other All

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